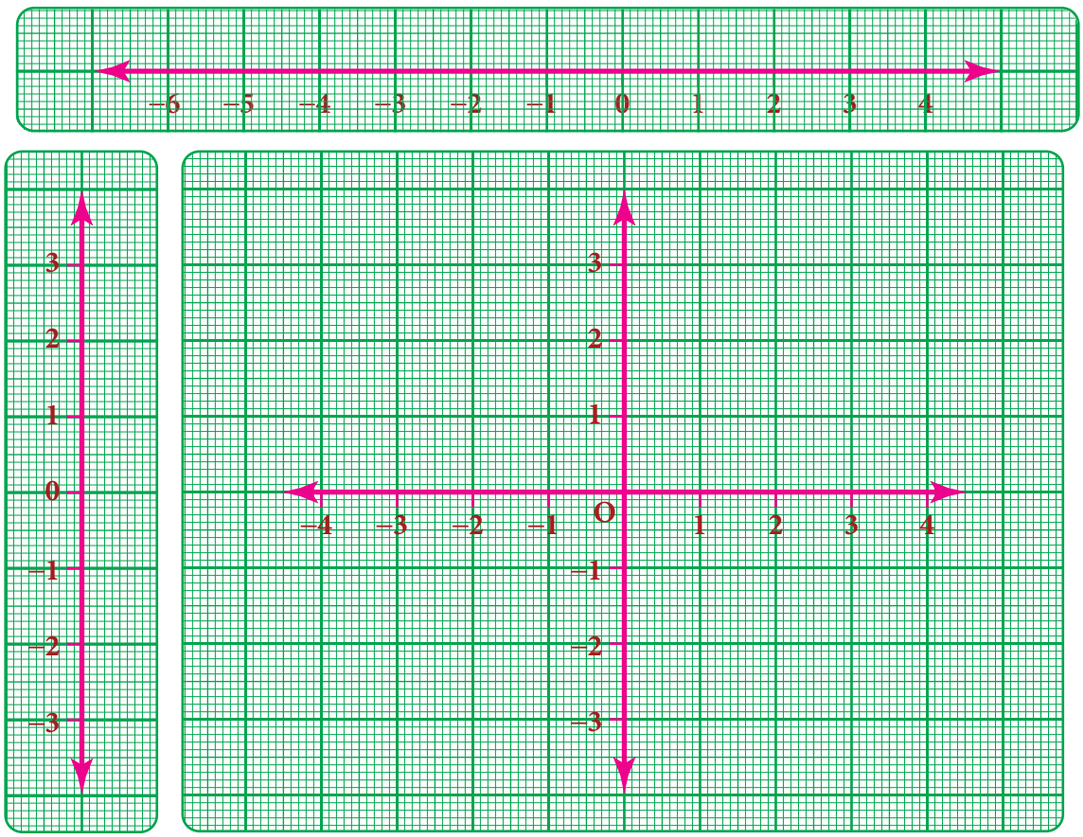


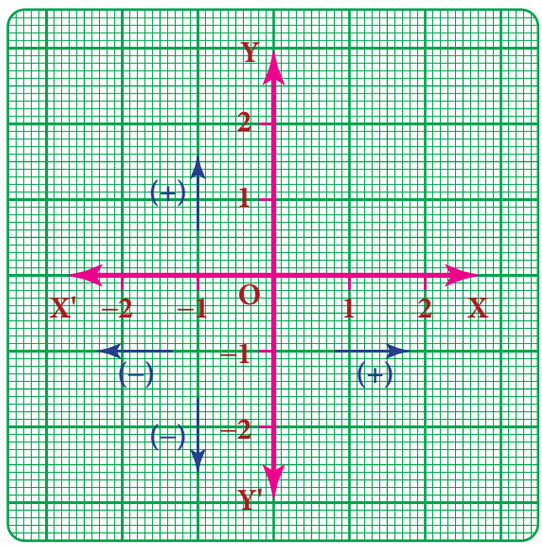
The intersecting point of the perpendicular lines 'O' represent the **origin (0, 0)**.

Cartesian system

Rene Descartes system of fixing a point with the help of two measurements, horizontal and vertical, is named as **Cartesian system,** in his honour. The horizontal line is named as XOX', called the X-axis. The vertical line is named as YOY', called the Y- axis. Both the axes are called **coordinate axes**. The plane containing the X axis and the Y axis is known as the coordinate plane or the **Cartesian plane.**

3.9.3 Signs in the graphs

1. X-coordinate of a point is positive along OX and negative along OX'
2. Y-coordinate of a point is positive along OY and negative along OY'



3.9.4 Ordered pairs

A point represents a position in a plane. A point is denoted by a pair (a,b) of two numbers 'a' and 'b' listed in a specific order in which 'a' represents the distance along the X-axis and 'b' represents the distance along the Y-axis. It is called an **ordered pair (*a*,*b*)**. It helps us to locate *precisely* a point in the plane. Each point can be exactly identified by a pair of numbers. It is also clear that the point (*b*,*a*)

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